WEATHER ELEMENTS / INDIA JET STREAM-WESTERN DISTURBANCES

WEATHER CAUSATION CHAIN → HEAT-BUDGET LEDGER → STABILITY PROFILES → WIND-FORCE TRIANGLE → PLANETARY CIRCULATION RAIL → HUMIDITY-TO-CLOUD CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

★ PRIMARY CORE FLOW · ◆ SUBORDINATE EXTRA · ▲ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 WEATHER CAUSATION CHAIN

WEATHER CAUSATION CHAIN · SURFACE AND ATMOSPHERIC HEATING · TEMPERATURE GRADIENT · PRESSURE GRADIENT · MOISTURE TRANSPORT AND CONVERGENCE · LIFT + SATURATION · CLOUD AND PRECIPITATION

INSOLATION → surface and atmospheric heating → TEMPERATURE GRADIENT → pressure gradient → WIND → moisture transport and convergence → LIFT + SATURATION → cloud and precipitation

DEFINITION / COMPONENTS
- INSOLATION → surface and atmospheric heating
- TEMPERATURE GRADIENT → pressure gradient

TRIGGER / MECHANISM
- WIND → moisture transport and convergence
- LIFT + SATURATION → cloud and precipitation

EFFECT / INTERVENTION
- INSOLATION → surface and atmospheric heating
- TEMPERATURE GRADIENT → pressure gradient

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WIND → moisture transport and convergence; LIFT + SATURATION → cloud and precipitation.

01

STAGE 01 HEAT-BUDGET LEDGER

HEAT-BUDGET LEDGER · INCOMING SHORTWAVE · REFLECTED + ABSORBED ENERGY · LONGWAVE + SENSIBLE + LATENT HEAT · TEMPERATURE AND CIRCULATION RESPONSE

SUN → incoming shortwave → SURFACE → reflected + absorbed energy → EARTH → longwave + sensible + latent heat → BALANCE → temperature and circulation response

DEFINITION / COMPONENTS
- SUN → incoming shortwave
- SURFACE → reflected + absorbed energy

TRIGGER / MECHANISM
- EARTH → longwave + sensible + latent heat
- BALANCE → temperature and circulation response

EFFECT / INTERVENTION
- SUN → incoming shortwave
- SURFACE → reflected + absorbed energy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EARTH → longwave + sensible + latent heat.

02

STAGE 02 STABILITY PROFILES

STABILITY PROFILES · NORMAL TENDENCY · TEMPERATURE FALLS UPWARD · TEMPERATURE RISES THROUGH LAYER · STABLE LAYER · VERTICAL MIXING SUPPRESSED · SURFACE INVERSION · FOG AND POLLUTION TRAPPING

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
NORMAL TENDENCY → temperature falls upward	STABLE LAYER → vertical mixing suppressed	NORMAL TENDENCY → temperature falls upward
INVERSION → temperature rises through layer	SURFACE INVERSION → fog and pollution trapping	INVERSION → temperature rises through layer

DEFINITION / COMPONENTS
- NORMAL TENDENCY → temperature falls upward
- INVERSION → temperature rises through layer

TRIGGER / MECHANISM
- STABLE LAYER → vertical mixing suppressed
- SURFACE INVERSION → fog and pollution trapping

EFFECT / INTERVENTION
- NORMAL TENDENCY → temperature falls upward
- INVERSION → temperature rises through layer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STABLE LAYER → vertical mixing suppressed; SURFACE INVERSION → fog and pollution trapping.

03

STAGE 03 WIND-FORCE TRIANGLE

WIND-FORCE TRIANGLE · PRESSURE GRADIENT · STARTS ACCELERATION · RIGHT NORTH · LEFT SOUTH · SLOWS AND CROSSES ISOBARS · NEAR-GEOSTROPHIC PARALLEL FLOW

PRESSURE GRADIENT → starts acceleration → CORIOLIS → right north / left south → FRICTION → slows and crosses isobars → ALOFT → near-geostrophic parallel flow

DEFINITION / COMPONENTS
- PRESSURE GRADIENT → starts acceleration
- CORIOLIS → right north / left south

TRIGGER / MECHANISM
- FRICTION → slows and crosses isobars
- ALOFT → near-geostrophic parallel flow

EFFECT / INTERVENTION
- PRESSURE GRADIENT → starts acceleration
- CORIOLIS → right north / left south

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FRICTION → slows and crosses isobars; ALOFT → near-geostrophic parallel flow.

04

STAGE 04 PLANETARY CIRCULATION RAIL

PLANETARY CIRCULATION RAIL · EQUATORIAL LOW · CONVERGENCE AND ASCENT · SUBTROPICAL HIGH · DESCENT AND TRADES · SUBPOLAR LOW · WESTERLY-POLAR INTERACTION · POLAR HIGH

EQUATORIAL LOW → convergence and ascent

SUBTROPICAL HIGH → descent and trades → SUBPOLAR LOW → westerly-polar interaction → POLAR HIGH → cold outflow; belts migrate

SYSTEM / DEFINITION
- EQUATORIAL LOW → convergence and ascent
- SUBTROPICAL HIGH → descent and trades

PROCESS / MECHANISM
- SUBPOLAR LOW → westerly-polar interaction
- POLAR HIGH → cold outflow; belts migrate

SPATIAL / INDIA PATTERN
- EQUATORIAL LOW → convergence and ascent
- SUBTROPICAL HIGH → descent and trades

HAZARD / LIMIT / RESPONSE
- SUBPOLAR LOW → westerly-polar interaction
- POLAR HIGH → cold outflow; belts migrate

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SUBPOLAR LOW → westerly-polar interaction; POLAR HIGH → cold outflow; belts migrate.

05

STAGE 05 HUMIDITY-TO-CLOUD CHAIN

HUMIDITY-TO-CLOUD CHAIN · MOISTURE CONTENT + TEMPERATURE · RELATIVE HUMIDITY · COOLING TO SATURATION · DEW POINT REACHED · NUCLEI + LIFT · DROPLETS OR ICE CRYSTALS · GROWTH + FALL SPEED

MOISTURE CONTENT + TEMPERATURE → relative humidity → COOLING TO SATURATION → dew point reached → NUCLEI + LIFT → droplets or ice crystals → GROWTH + FALL SPEED → precipitation

DEFINITION / COMPONENTS
- MOISTURE CONTENT + TEMPERATURE → relative humidity
- COOLING TO SATURATION → dew point reached

TRIGGER / MECHANISM
- NUCLEI + LIFT → droplets or ice crystals
- GROWTH + FALL SPEED → precipitation

EFFECT / INTERVENTION
- MOISTURE CONTENT + TEMPERATURE → relative humidity
- COOLING TO SATURATION → dew point reached

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NUCLEI + LIFT → droplets or ice crystals.

06

STAGE 06 RAINFALL COMPARISON

RAINFALL COMPARISON · BUOYANT THERMAL ASCENT · FORCED RELIEF ASCENT · CONVERGENT DYNAMIC ASCENT · REAL EVENT · MECHANISMS MAY COMBINE

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
CONVECTIONAL → buoyant thermal ascent	FRONTAL / CYCLONIC → convergent dynamic ascent	CONVECTIONAL → buoyant thermal ascent
OROGRAPHIC → forced relief ascent	REAL EVENT → mechanisms may combine	OROGRAPHIC → forced relief ascent

DEFINITION / COMPONENTS
- CONVECTIONAL → buoyant thermal ascent
- OROGRAPHIC → forced relief ascent

TRIGGER / MECHANISM
- FRONTAL / CYCLONIC → convergent dynamic ascent
- REAL EVENT → mechanisms may combine

EFFECT / INTERVENTION
- CONVECTIONAL → buoyant thermal ascent
- OROGRAPHIC → forced relief ascent

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FRONTAL / CYCLONIC → convergent dynamic ascent; REAL EVENT → mechanisms may combine.

07

STAGE 07 TROPOSPHERE SECTION

TROPOSPHERE SECTION · MOISTURE, FRICTION AND HEATING · MID LEVELS · CLOUDS, TROUGHS AND STABILITY · UPPER TROPOSPHERE · JETS AND DIVERGENCE · TROPICS DEEPER · POLES SHALLOWER

SURFACE → moisture, friction and heating

MID LEVELS → clouds, troughs and stability → UPPER TROPOSPHERE → jets and divergence → TROPICS DEEPER / POLES SHALLOWER

SYSTEM / DEFINITION
- SURFACE → moisture, friction and heating
- MID LEVELS → clouds, troughs and stability

PROCESS / MECHANISM
- UPPER TROPOSPHERE → jets and divergence
- TROPICS DEEPER / POLES SHALLOWER

SPATIAL / INDIA PATTERN
- SURFACE → moisture, friction and heating
- MID LEVELS → clouds, troughs and stability

HAZARD / LIMIT / RESPONSE
- UPPER TROPOSPHERE → jets and divergence
- TROPICS DEEPER / POLES SHALLOWER

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UPPER TROPOSPHERE → jets and divergence; TROPICS DEEPER / POLES SHALLOWER.

08

STAGE 08 JET-STREAM ENGINE

JET-STREAM ENGINE · HORIZONTAL TEMPERATURE GRADIENT · PRESSURE GRADIENT ALOFT · CORIOLIS BALANCE · STRONG NEAR-ZONAL WIND · ROSSBY WAVES · RIDGES, TROUGHS AND STEERING

HORIZONTAL TEMPERATURE GRADIENT → HEIGHT / PRESSURE GRADIENT ALOFT → CORIOLIS BALANCE → strong near-zonal wind → ROSSBY WAVES → ridges, troughs and steering

DEFINITION / COMPONENTS
- HORIZONTAL TEMPERATURE GRADIENT
- HEIGHT / PRESSURE GRADIENT ALOFT

TRIGGER / MECHANISM
- CORIOLIS BALANCE → strong near-zonal wind
- ROSSBY WAVES → ridges, troughs and steering

EFFECT / INTERVENTION
- HORIZONTAL TEMPERATURE GRADIENT
- HEIGHT / PRESSURE GRADIENT ALOFT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CORIOLIS BALANCE → strong near-zonal wind; ROSSBY WAVES → ridges, troughs and steering.

09

STAGE 09 INDIA SEASONAL JETS

INDIA SEASONAL JETS · STWJ SOUTH OF HIMALAYA SECTOR · STEERS WESTERN DISTURBANCES EAST · TROPICAL EASTERLY JET DEVELOPS · JETS ARE PARTS OF WIDER CIRCULATION

WINTER → STWJ south of Himalaya sector

STWJ → steers western disturbances east → SUMMER → tropical easterly jet develops → RULE → jets are parts of wider circulation

SYSTEM / DEFINITION
- WINTER → STWJ south of Himalaya sector
- STWJ → steers western disturbances east

PROCESS / MECHANISM
- SUMMER → tropical easterly jet develops
- RULE → jets are parts of wider circulation

SPATIAL / INDIA PATTERN
- WINTER → STWJ south of Himalaya sector
- STWJ → steers western disturbances east

HAZARD / LIMIT / RESPONSE
- SUMMER → tropical easterly jet develops
- RULE → jets are parts of wider circulation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → jets are parts of wider circulation.

10

STAGE 10 WESTERN-DISTURBANCE CHAIN

WESTERN-DISTURBANCE CHAIN · UPPER TROUGH · EASTWARD WESTERLY MOTION · MOISTURE + DYNAMICAL LIFT · CLOUD AND RAIN/SNOW · HIMALAYA + PLAINS · TERRAIN-SPECIFIC IMPACTS · BENEFICIAL OR DAMAGING BY TIMING/INTENSITY

UPPER TROUGH / LOW → eastward westerly motion → MOISTURE + DYNAMICAL LIFT → cloud and rain/snow → HIMALAYA + PLAINS → terrain-specific impacts → OUTCOME → beneficial or damaging by timing/intensity

DEFINITION / COMPONENTS
- UPPER TROUGH / LOW → eastward westerly motion
- MOISTURE + DYNAMICAL LIFT → cloud and rain/snow

TRIGGER / MECHANISM
- HIMALAYA + PLAINS → terrain-specific impacts
- OUTCOME → beneficial or damaging by timing/intensity

EFFECT / INTERVENTION
- UPPER TROUGH / LOW → eastward westerly motion
- MOISTURE + DYNAMICAL LIFT → cloud and rain/snow

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MOISTURE + DYNAMICAL LIFT → cloud and rain/snow.

11

STAGE 11 FORECAST EVIDENCE LADDER

FORECAST EVIDENCE LADDER · PILOT BALLOON · UPPER-AIR STATE · SATELLITE + RADAR · CLOUD AND PRECIPITATION · NWP ENSEMBLE · SCENARIOS AND UNCERTAINTY · ACTION; LATER OBSERVATIONS VERIFY OUTCOME

RADIOSONDE / PILOT BALLOON → upper-air state → SATELLITE + RADAR → cloud and precipitation → NWP ENSEMBLE → scenarios and uncertainty → WARNING → action; later observations verify outcome

DEFINITION / COMPONENTS
- RADIOSONDE / PILOT BALLOON → upper-air state
- SATELLITE + RADAR → cloud and precipitation

TRIGGER / MECHANISM
- NWP ENSEMBLE → scenarios and uncertainty
- WARNING → action; later observations verify outcome

EFFECT / INTERVENTION
- RADIOSONDE / PILOT BALLOON → upper-air state
- SATELLITE + RADAR → cloud and precipitation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NWP ENSEMBLE → scenarios and uncertainty; WARNING → action; later observations verify outcome.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · JET STREAMS ARE FAST, NARROW UPPER-TROPOSPHERIC WIND BANDS. · STWJ LIES SOUTH OF THE HIMALAYAS IN WINTER AND · NORTHWARD WITHDRAWAL OF STWJ IS A PRE-CONDITION FOR SW · TEJ DEVELOPS AS PART OF THE SUMMER MONSOON'S UPPER-AIR · WESTERN DISTURBANCES ARE EXTRA-TROPICAL, NON-MONSOONAL LOW-PRESSURE SYSTEMS FROM THE · WDS BRING WINTER/EARLY-SPRING RAIN TO PUNJAB, HARYANA, DELHI AND · THEY ARE VITAL FOR RABI CROPS SUCH AS WHEAT

OPTIONAL PROCESS DEPTH
- Jet streams are fast, narrow upper-tropospheric wind bands.
- STWJ lies south of the Himalayas in winter and acts as the conveyor for Western Disturbances.
- Northward withdrawal of STWJ is a pre-condition for SW monsoon onset.
- TEJ develops as part of the summer monsoon's upper-air circulation. Tibetan/Asian heating matters.

DATA / INSTITUTION LIMIT
- Western Disturbances are extra-tropical, non-monsoonal low-pressure systems from the Mediterranean moving east via Iran-Afghanistan-Pakistan.
- WDs bring winter/early-spring rain to Punjab, Haryana, Delhi and UP, and snow to J&K, HP and Uttarakhand
- they are vital for Rabi crops such as wheat, mustard and gram.
- In May–June, intense heating over north-west India supports the hot, dry Loo. Its timing and strength

CONTEMPORARY PARALLEL
- Altitude lowers temperature by about 6.5 deg C/km in the notes
- the Himalayan barrier gives India warmer winters than comparable latitudes by blocking Central Asian cold air.
- STWJ steers Western Disturbances in winter.
- STWJ's northward shift is essential for SW monsoon onset.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels